Software Module

Technologies

- ESP32 Firmware
- Bluetooth Communication
- Mobile Application
- Machine Learning Model
- Data Visualization Dashboard

Software Working

The ESP32 collects load sensor data and transmits it to the mobile application using Bluetooth communication.

The mobile application stores and analyzes the received data. A machine learning algorithm evaluates rehabilitation progress based on user information and walking patterns.

The application provides:

- Real-time weight monitoring
- Progress tracking
- Rehabilitation status visualization
- Historical data analysis
- Alert notifications

Features

- Bluetooth Connectivity
- Real-Time Monitoring
- Recovery Progress Dashboard
- Data Logging
- Machine Learning Based Analysis
- Doctor/Caregiver Monitoring (Future)

Future Software Enhancements

- Android Application
- Doctor Dashboard
- Cloud Storage
- Hospital Management Integration
- AI-Based Recovery Prediction
- Remote Monitoring Platform